

SHANMUGA VADIVEL NACHIMUTHU

AI-First QA Engineer | SDET

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Coimbatore, Tamil Nadu

SUMMARY

Experienced AI-first QA Engineer / SDET with 8+ years of experience specializing in intelligent test automation, quality engineering, and test automation framework design, using Playwright, Selenium, and TestCafe with strong programming skills in Java and TypeScript. Proficient in AI-driven testing, Generative AI, LLM testing, Agentic AI, and shift-left testing within CI/CD pipelines (Jenkins, GitLab), along with API testing (REST, Postman) and BDD (Cucumber). Successfully improved automation execution pass rate from 35% to 85%+ and implemented AI-driven self-healing automation with LLM integration, significantly enhancing execution stability and reducing manual effort. Hands-on experience with RAG-based testing, AI model evaluation (DeepEval, RAGAs), and local LLM integration, enabling scalable, cost-optimized automation solutions and faster, high-quality software delivery in Agile environments.

EXPERIENCE

Quality Engineering Senior Analyst

Accenture Solutions Private Limited

04/2022 - Present Coimbatore, India

- Architected a scalable test automation framework using Selenium, Playwright, TestCafe, TestNG, and Cucumber (BDD), reducing regression execution time by ~40% and improving maintainability.
- Automated over 400 UI, API, and regression test scenarios, cutting manual testing effort by approximately 65–70% per sprint.
- Established API automation solutions using REST Assured, Postman, and Bruno, achieving over 90% API validation coverage across critical services.
- Orchestrated CI/CD-driven automation pipelines with Jenkins and GitLab, reducing feedback cycle time from approximately 6 hours to under 2 hours, making it about 65% faster.
- Strengthened release quality by reducing production defects by approximately 35–40%, leveraging enhanced regression coverage and shift-left testing practices.
- Pioneered AI-first testing strategies using Playwright and LLM-based prompt engineering, improving test generation efficiency by approximately 40% and enabling automated test case creation with JIRA integration via AI Workbench, reducing effort by about 50%.
- Conducted architecture-level framework validation using AI-driven prompt instructions and accelerated peer review cycles by approximately 25–30% through AI-assisted code analysis.
- Enabled Agentic AI-driven test execution, incorporating self-healing mechanisms that dynamically resolve locator failures and improve execution stability by approximately 35%, with persistent storage reducing repeated LLM calls by approximately 20–25%.
- Integrated local LLM models (Ollama) and applied RAG-based validation, DeepEval, and RAGAs, reducing LLM costs by approximately 30–40% while improving AI system validation accuracy.
- Guided and mentored more than 5 QA engineers, improving onboarding efficiency by approximately 30% through structured knowledge transfer, reusable frameworks, and best practices.

Associate Software Engineer

Bosch Global Software Technologies

03/2020 - 04/2022 Bangalore, India

- Engineered and maintained 300+ UI automation scripts using Selenium, Protractor, and TypeScript, reducing manual regression effort by approximately 60–65%.
- Expanded automation coverage by approximately 45–50% across critical business workflows, strengthening release confidence.
- Accelerated test execution by approximately 35–40% through parallel execution enabled via Jenkins CI/CD pipelines.
- Conducted API validation using Postman, improving backend coverage by approximately 30% and detecting integration defects early.
- Elevated automation stability from 35% to over 85% execution success rate by eliminating flaky failures through synchronization and script optimization.
- Contributed to BDD framework adoption using Cucumber, improving collaboration between QA, development, and business teams.
- Supported Agile ceremonies, enhancing test estimation, sprint planning, and delivery quality.
- Optimized test data handling and reusable components, reducing maintenance overhead by approximately 25%.
- Coached two junior QA engineers, improving onboarding speed and reducing ramp-up time by approximately 25%.

SKILLS

Automation & Frameworks

Selenium	Playwright	TestCafe	TestNG
Cucumber (BDD)			

Programming & Scripting

Java	TypeScript	SQL
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API Testing and Tools

RESTAssured	Postman	Bruno
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AI & Advanced Testing

AI-driven Testing	Generative AI	LLM Testing
Agentic AI	AI Model Evaluation	DeepEval
RAGA		

CI/CD & DevOps

Jenkins	GitLab	Maven	Gradle
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Test Management & Process

JIRA	HP ALM	Agile
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KEY ACHIEVEMENTS



OTA Award - BOSCH

Increased automation execution success rate from 35% to 85%+ by stabilizing tests and improving framework reliability, ensuring consistent regression outcomes.



ACE Award - Accenture

Rapidly upskilled in project-critical technologies, including Playwright-based automation and AI-driven testing approaches, enabling successful validation and on-time delivery of a critical deployment under aggressive timelines.

CERTIFICATION

ISTQB – CTFL (Certified Tester Foundation Level)

COURSES

Learn Agentic AI – Build Multi-Agent Automation Workflows

AI-Driven Test Automation: Playwright, Selenium, LLMs & More

2026-Using Gen AI & AI Agent in Software Automation Testing

Test AI & LLM App with DeepEval, RAGAs & more using Ollama

EXPERIENCE

Project Engineer

Wipro Technologies Private Limited

10/2017 - 03/2020 Bangalore, India

- Engineered 150+ automation scripts using Selenium WebDriver and TestNG, supporting regression and functional validation cycles.
- Assisted in building foundational automation framework components, improving reusability and reducing script duplication by approximately 25%.
- Reduced manual repetitive testing effort by approximately 40% by automating high-frequency test scenarios.
- Administered defect lifecycle management using HP ALM, improving defect traceability and reducing resolution turnaround time.
- Performed functional, regression, smoke, and sanity testing, ensuring stable builds and improved test coverage across releases.
- Collaborated with cross-functional teams in an Agile environment, enhancing test planning accuracy and on-time delivery.
- Streamlined test data preparation and environment validation, reducing configuration effort by approximately 20%.

COURSES

Java Data Structures & Algorithms + LEETCODE Exercises

EDUCATION

Bachelor of Engineering (B.E)

K.S.Rangasamy College of Technology

2013 - 2017

LANGUAGES

Tamil

Native

English

Advanced